

PHYSICS 77/88 Final Project: Simulating the 2D Ising Model with the Metropolis-Hastings Algorithm—Critical Phenomena

Alexander Carleton Halpin^{1,*} and Eden Michael Huang^{1,†}

¹*Department of Physics, University of California, Berkeley*

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Overview: There are many systems in nature that can experience phase transitions, such as water boiling. Certain quantities in these systems follow power laws governed by critical exponents near the phase transition. Many seemingly unrelated systems share the same critical exponents. We explore how systems that exhibit criticality behave near phase transitions using the two-dimensional Ising model, the simplest system that undergoes a continuous phase transition at a non-zero temperature. All of the data used is from our Ising model simulation, which we sample to observe how it changes as time progresses and equilibrates at different temperatures. We used our simulated data to estimate critical exponents. Our estimated critical exponents are very close to the analytically calculated values. We were limited by computational resources, so some aspects of our Ising model simulation were not idealized.

I. INTRODUCTION AND PHYSICS MOTIVATION

Problem statement. Many physical systems undergo phase transitions: dramatic changes in collective behavior that occur at a precise critical temperature. Familiar examples include water boiling and iron losing its magnetization above the Curie point. The 2D Ising model is a two-dimensional lattice of “spins” that can either point up or down, and interact only with their nearest neighbors, preferring to align. There exists a critical temperature at which the system transitions from an ordered phase to a disordered phase. We answer whether a computationally-feasible finite-lattice Monte Carlo simulation can recover the universal critical exponents, which describe how fast certain quantities diverge as the system approaches criticality.

Why it matters. Understanding phase transitions in the 2D Ising model is important because many physical systems undergo transitions governed by the same critical exponents, a phenomenon called universality [1]. As systems approach criticality, the microscopic details of the lattice become irrelevant [2]. Consequently, the critical exponents of the 2D Ising model also apply to seemingly unrelated systems in the same universality class, defined by dimensionality ($d = 2$) and symmetry of the order parameter (discrete $\mathbb{Z}_2$ or “up-down” symmetry in the case of the 2D Ising model) [3].

Why/where computational techniques come in. Empirical measurement of critical exponents requires extremely precise control and measurement of the system, which is nearly impossible in physical experiments, but very feasible with NumPy. Computational techniques also come in when averaging results across trials, running regressions on data to estimate critical exponents

and calculating uncertainty. We demonstrate that relatively small-scale computational data can be used to accurately estimate the behavior of large-scale physical systems.

Contributions.

- Implemented the Metropolis–Hastings algorithm
- Built a “burn-in” diagnostic
- Simulated average magnetization at temperatures from $T = 0.5$ to $T = 4$
- Simulated average magnetization at different external magnetic field strengths ranging from $B = -0.5$ to $B = 0.5$ at the critical temperature
- Fitted curves to the average magnetization vs. temperatures and average magnetization vs. external magnetic field to estimate the critical exponents
- Calculated error bars on the critical exponents

Paper roadmap. Section II covers background and prior work; Section III describes the simulation setup; Section IV details the Metropolis–Hastings algorithm; Section V presents our results for β and δ ; Section VI discusses validation, uncertainty, and limitations; and Section VII describes reproducibility.

II. BACKGROUND AND RELATED WORK

Physics background. The Ising model assigns a spin value $\sigma_i \in \{-1, +1\}$ to each site on a lattice, where spins interact with their nearest neighbors and prefer to align when the coupling constant $J > 0$. The total energy of any configuration is given by:

$$E = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - B \sum_i \sigma_i \quad (1)$$

* ahalpin@berkeley.edu

† edenhuangg@berkeley.edu

where the first sum runs over all nearest-neighbor pairs and B is an external magnetic field. At any temperature T , the probability of the system being in a particular configuration is given by the Boltzmann distribution $p \propto e^{-E/T}$, meaning low-energy (aligned) configurations are exponentially more likely at low temperatures, and all configurations become roughly equally likely at high temperatures.

The central quantity we track is the order parameter, average magnetization:

$$m = \frac{1}{N^2} \sum_i \sigma_i \quad (2)$$

where N^2 represents the number of spins in the lattice. Near the critical temperature T_c , the magnetization vanishes according to a power law controlled by the critical exponent β , and at exactly T_c the response to an external field is controlled by the exponent δ :

$$m(\tau) \sim |\tau|^\beta \quad (\tau < 0), \quad m(T_c, B) \sim \begin{cases} B^{1/\delta}, & B > 0 \\ (-B)^{1/\delta}, & B < 0 \end{cases} \quad (3)$$

where $\tau = \frac{(T-T_c)}{T_c}$.

Prior work. The Ising model has a long analytic history that directly motivates and validates our computational approach. In 1925, Ernest Ising (the namesake of the Ising model) solved the one-dimensional model exactly, showing it has no phase transition at any finite temperature [4]. This result established the model as a precise mathematical object and motivated the search for richer behavior in higher dimensions. Onsager (1944) solved the 2D model exactly in zero external field [1]. He analytically derived the critical temperature $T_c = \frac{2J}{\ln(1+\sqrt{2})} \approx 2.269 \cdot J$ and showed that there is a phase transition at T_c . We use this value of T_c extensively in our simulations. Yang (1952) extended Onsager's result [5] by computing critical exponents analytically, establishing $\beta = 1/8$ and $\delta = 15$. We use these values to validate our model. Metropolis et al. (1953) introduced [6] the Monte Carlo algorithm we implement. We use the idea of accepting energy-lowering moves always, and energy-raising moves with Boltzmann probability in our implementation. Hastings (1970) generalized [7] the Metropolis algorithm to arbitrary proposal distributions, establishing the theoretical foundation for the broader class of Markov Chain Monte Carlo (MCMC) methods. His work guarantees that our sampling procedure converges to the correct Boltzmann distribution given sufficient sweeps. This is the theory behind the "burn-in" phase of our implementation. These papers provide the framework for our simulation and quantitative methods to validate our simulation.

More work has been done to establish the universality of the Ising model, which motivated our project. Kadanoff (1966) showed that the Ising model near the critical temperature looks the same at different length scales, a concept called block-spin scaling [8]. Wilson

(1971) transformed this concept into the Renormalization Group, the idea that different microscopic systems flow toward the same "fixed point" under scale transformations [9]. This is why universality exists. Belavin, Polyakov, and Zamolodchikov (1984) established Conformal Field Theory (CFT), which shows that there is conformal invariance, a higher level of symmetry, at the critical temperature, which simplifies finding critical exponents and further classifies different universality classes [10]. CFT also showed that the 2D Ising model is the simplest model in its universality class. More modern papers provide more evidence and clarity needed to prove that the Ising model's behavior is only dependent on its dimensionality the symmetry of its order parameter [2][3].

III. DATA AND EXPERIMENTAL/SIMULATION SETUP

Data sources. All data in this project are entirely simulated, no external datasets, observations, or experimental measurements are used. The raw output of each simulation run is generated fresh by the Metropolis-Hastings algorithm and stored as NumPy arrays in compressed files (`ising_data-1.npz`, `ising_delta_data (2).npz` ~3 KB) containing the following fields:

Field	Description
<code>T_range</code>	Range of temperatures tested
<code>mean_M_arr</code>	Mean magnetization per spin at each T
<code>std_M_arr</code>	Trial-to-trial standard deviation of magnetization
<code>mean_E_arr</code>	Mean energy per spin at each T
<code>std_E_arr</code>	Trial-to-trial standard deviation of energy
<code>mean_Cv</code>	Mean heat capacity per spin at each T
<code>mean_chi</code>	Mean magnetic susceptibility at each T

TABLE I. Fields stored in `ising_data.npz`.

Field	Description
<code>B_range</code>	Range of external magnetic field strengths tested
<code>mean_M_B</code>	Mean average magnetization at each B-field
<code>std_error_M_B</code>	Standard deviation of average magnetization

TABLE II. Fields stored in `ising_delta_data (2).npz`.

Setup and Preprocessing. Rather than filtering or normalizing raw experimental data, our preprocessing consists of the choices made before the simulation begins.

Lattice initialization. Each run begins with a biased random lattice: each spin is set to +1 with probability 0.6 and -1 with probability 0.4. When running a simulation with a negative external magnetic field, these initialization probabilities are reversed. This mild bias is chosen to help the system reach equilibrium faster at low temperatures without artificially biasing measurements at the critical point.

Burn-in (equilibration). The first 250 sweeps of every run are discarded entirely and never enter the measurements. A sweep consists of $N^2 = 2500$ individual spin-flip attempts. This burn-in period allows the lattice to forget its initial condition and settle into a representative equilibrium state. The choice of 250 sweeps was validated by tracking $|m|$ vs. sweep number across 5 independent trials at 7 temperatures spanning $T = 0.5$ to $T = 5.0$ (Fig. 1). In all cases magnetization stabilized well before sweep 250. Note that this validation range extends to $T = 5.0$, whereas the main simulation grid runs only to $T = 4.0$.

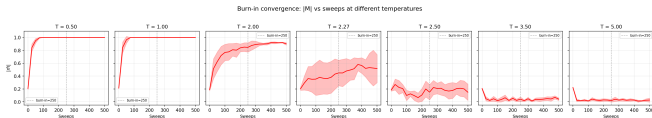


FIG. 1. Burn-in convergence: $|M|$ vs. sweeps at seven representative temperatures. The dashed vertical line marks the burn-in cutoff at sweep 250. Shaded bands show the standard deviation across 5 independent trials.

Temperature and magnetic field ranges. Temperatures are sampled non-uniformly: 10 points in the low-temperature region $T \in [0.5, 2.0]$, 20 points in the critical region $T \in [2.0, 2.5]$ (denser, to resolve the phase transition), and 10 points in the high-temperature region $T \in [2.5, 4.0]$, giving 40 temperatures in total. External magnetic field strengths are sampled around $B = 0$: 41 points in the near-zero field region $B \in [-0.5, 0.5]$.

Measurement and averaging. After burn-in, 500 measurement sweeps are collected per trial, recording instantaneous magnetization at each sweep. These 500 samples are averaged to give a single estimate per trial. Three independent trials are run at each temperature, and their mean and standard deviation are recorded; the standard deviation across trials serves as the uncertainty (error bar) on each data point.

IV. METHODS

Computational pipeline. The simulation proceeds in six steps:

1. **Set** a temperature and external magnetic field strength at which the simulation will run.
2. **Initialize** a 50×50 lattice with spins (-1 or $+1$) drawn from a biased random distribution.
3. **Equilibrate** by running 250 burn-in sweeps, discarding all measurements.
4. **Measure** by running 500 sweeps, recording average magnetization after each sweep.
5. **Repeat** steps 1–3 for 3 independent trials.

6. **Average** across trials to get the mean and standard deviation of average magnetization at each the set temperature and external magnetic field.
7. **Repeat** steps 1–6 with a different temperature or external magnetic field strength.
8. **Fit** power-law models to the aggregated data via a log-log linear regression to extract critical exponents β and δ .

Numerical. The core algorithm is the Metropolis–Hastings Monte Carlo method. At each step, a spin is selected at random. The change in energy ΔE of flipping that spin computed from its four nearest neighbors under periodic boundary conditions. The spin flip is accepted with probability

$$P(\Delta E) = \begin{cases} 1, & \Delta E \leq 0 \\ e^{-\Delta E \beta}, & \Delta E > 0 \end{cases} \quad (4)$$

where $\beta = 1/k_B T$. We chose units so that $k_B = 1$. Critical exponents are extracted by weighted least-squares linear regressions (`scipy.optimize.curve_fit`) on the log-log transformed power-law tails of $|m|$ vs. τ and $|m|$ vs. B at T_c .

Key equations. The total energy of the lattice is

$$E = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - B \sum_i \sigma_i \quad (5)$$

where J is the coupling constant (we always set $J = 1$), B is the external field, and the first sum runs over nearest-neighbor pairs. The reduced temperature is

$$\tau = \frac{T - T_c}{T_c} \quad (6)$$

Near the critical point, the order parameter scales as

$$|m| \sim |\tau|^\beta \quad (7)$$

The external field dependence of magnetization at T_c scales as

$$|m| \sim |B|^{1/\delta} \quad (8)$$

Log-log transformations of (7) and (8) give

$$\ln |m| = C + \beta \ln |\tau|, \quad \ln |m| = C + \frac{1}{\delta} \ln |B| \quad (9)$$

where C is a constant that we disregard. (9) can be used to find β and δ from the slope of the log-log plots. The exact Onsager–Yang values for the 2D Ising model are $\beta = 1/8$ and $\delta = 15$ [1, 5].

Hyperparameters. The key simulation parameters are: lattice size $N = 50$, burn-in sweeps = 250, measurement sweeps = 500, independent trials per temperature = 3, and coupling constant $J = 1$. The burn-in length was chosen by visual inspection of $|m|$ vs. sweep curves (Fig. 1); all other values were set by the computational feasibility constraints of running on a laptop CPU.

Code snippet. Listing 1 shows the core Metropolis sweep and Listing 2 shows the vectorized energy calculation.

Listing 1. Metropolis sweep implementing the Ising model in NumPy.

```
import numpy as np
N = 50
lattice = np.random.choice([-1, 1], size=(N, N))

def metropolis_sweep(lattice, T, J=1.0, B=0.0):
    Lx, Ly = lattice.shape
    beta = 1.0 / T # inverse temperature (kB = 1)
    for _ in range(Lx * Ly):
        i = np.random.randint(0, Lx)
        j = np.random.randint(0, Ly)
        spin = lattice[i, j]
        # periodic boundary conditions
        neighbours = (
            lattice[(i - 1) % Lx, j] + lattice[(i + 1) % Lx, j] +
            lattice[i, (j - 1) % Ly] + lattice[i, (j + 1) % Ly]
        )
        dE = 2.0 * spin * (J * neighbours + B)
        if dE <= 0 or np.random.random() < np.exp(-beta * dE):
            lattice[i, j] *= -1
    return lattice
```

Listing 2. Vectorized total energy of the Ising lattice.

```
import numpy as np
N = 50
lattice = np.random.choice([-1, 1], size=(N, N))

def total_energy(lattice, J=1.0, B=0.0):
    E = -J * np.sum(
        lattice * np.roll(lattice, -1, axis=1) + #
        right neighbor
        lattice * np.roll(lattice, -1, axis=0) #
        below neighbor
    )
    E -= B * np.sum(lattice)
    return E
```

V. RESULTS

A. Result I: $\beta = 0.1286 \pm 0.0030$

Main findings. Fig. 2 shows the results from simulating how average magnetization changes at different temperatures, ranging from $T = 0.5$ to $T = 4$. The power law behavior of average magnetization can be seen as T approaches T_c from the left. Note that average magnetization still has a finite value slightly above T_c rather than being perfectly flat above T_c .

Fig. 3 shows the log-log transformation of the average magnetization vs. Temperature data in Fig. 2 for $T < T_c$. $\ln(m)$ is plotted against $\ln|\tau|$ where $\tau = \frac{T - T_c}{T_c}$.

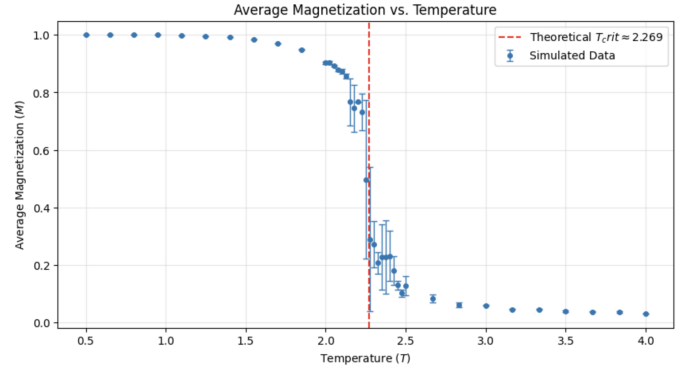


FIG. 2. Results from simulating average magnetization at different temperatures

The slope of the weighted least-squares regression line is 0.1286, with a standard deviation of 0.0030. This slope is our estimated value of β , so our simulation gives $\beta = 0.1286 \pm 0.0030$.

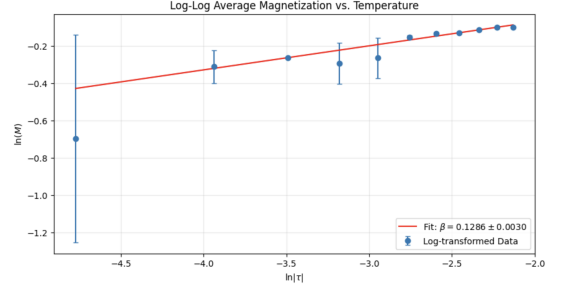


FIG. 3. Log-log transformation of average magnetization vs. temperature data, with a weighted least-squares regression line

B. Result II: $\delta = 17.3998 \pm 0.0687$

Fig. 4 shows the results from simulating how average magnetization changes at different external magnetic field strengths, ranging from $B = -0.5$ to $B = 0.5$. The power law behavior of average magnetization can be seen as B approaches zero from both sides. Note that all points are displayed with error bars, but standard deviation for all points other than $B = 0$ is on the order of 10^{-4} or less.

Fig. 5 and Fig 6. show the log-log transformation of the average magnetization vs. external magnetic field data from Fig. 4. Transformed data for $B < 0$ and $B > 0$ are shown in Fig. 5 and Fig. 6 respectively. Both plots also show the slopes of their respective weighted least-squares regression lines, which are equal to $1/\delta^\pm$. From the slopes in Fig. 5 and Fig. 6:

$$\delta^- = 17.1439 \pm 0.0687, \quad \delta^+ = 17.6556 \pm 0.0687, \quad (10)$$

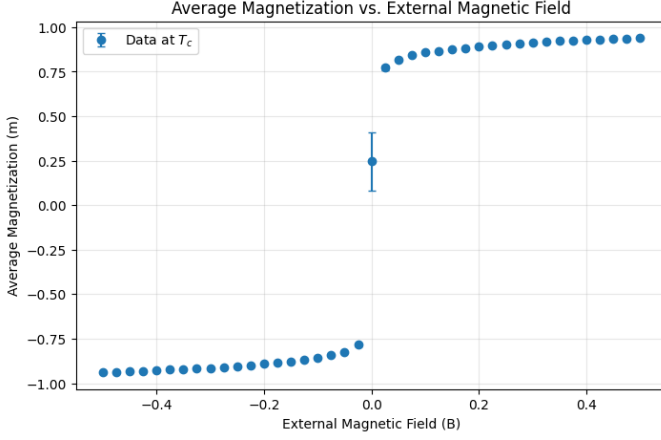


FIG. 4. Average magnetization vs. external magnetic field B at $T = T_c$.

Averaging these gives our simulation's final value of $\delta = 17.3998 \pm 0.0687$.

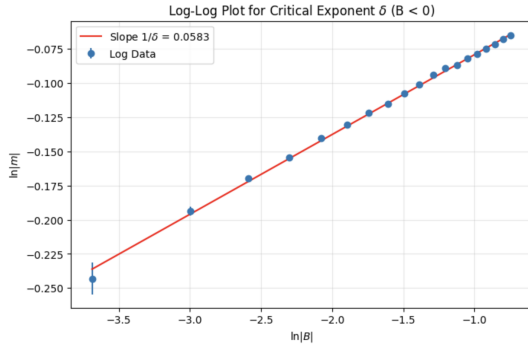


FIG. 5. Log-log transformation for $B < 0$, with weighted least-squares regression line giving $\delta^- = 17.1439 \pm 0.0687$.

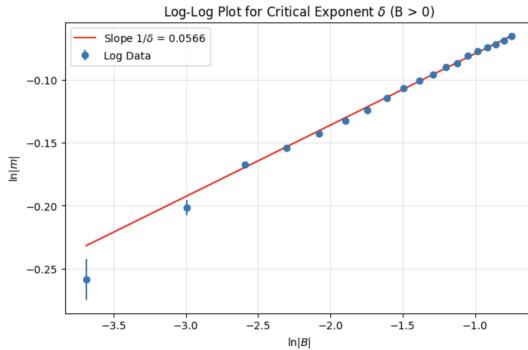


FIG. 6. Log-log transformation for $B > 0$, with weighted least-squares regression line giving $\delta^+ = 17.6556 \pm 0.0687$.

VI. DISCUSSIONS

Validations. Our simulated value of $\beta = 0.1286 \pm 0.0030$ is very close to the analytically calculated value of $\beta = 1/8 = 0.125$. This validates our model. Our simulated value of δ is quite close to, but still significantly off from the analytically calculated value of $\delta = 15$.

Uncertainty. $\frac{\hat{\beta} - \beta}{\sigma_\beta} = 1.2$ indicates that the difference between our estimation of β and its actual value is likely due to random variation in our simulation. On the other hand, $\frac{\hat{\delta} - \delta}{\sigma_\delta} = 34.9$, which is large, indicating the difference between our estimation of δ and its actual value is statistically significant.

Ablations. The finite size of our lattice is a possible reason why our estimation of δ is significantly different than the actual value. Theory on the Ising model says that as T approaches T_c , correlation length (the distance over which spins "talk") approaches ∞ [11]. The periodic boundary conditions in our simulated lattice mean that as T approaches T_c , spins become correlated with themselves. At the critical temperature, periodic boundary conditions make any spin flip less probable because each spin is correlated with itself. This means that order emerges slower in a finite lattice than in an infinite lattice. All of the simulated data for δ was set at T_c , and our overestimation of δ corresponds to an underestimation of how fast order emerges as $|B|$ is increased (remember that the slope of the regression line equals $1/\delta$). This raises the finite lattice size as a possible explanation of our simulation's overestimating δ .

Unfortunately, lattice size is a difficult variable to test because the number of spin flip checks needed for a full "sweep" scales $O(N^2)$, where N is the length of one side of the square lattice. From some preliminary simulations, it seems like the number of $N \cdot N$ sweeps required to equilibrate the model increases as the size of the model increases. Due to limitations on computational resources, it was not feasible for us to run simulations to see how our estimation of δ changes with lattice size.

Another key parameter is the biasing of the initial conditions. With unbiased initial conditions, the model takes much longer to equilibrate, and it is somewhat likely that the model stays at a near-zero average magnetization, even after many sweeps at a low temperature. We chose to bias our initial conditions so that the burn-in phase of our simulations could be shorter. The trade-off here is that biased initial conditions could make critically less "sharp," as seen in Fig 2. However, our relatively accurate estimation of β indicates that biased initial conditions are likely not problematic. We ran another simulation for δ with unbiased initial conditions and still overestimated the actual value by about the same amount.

Limitations. Our finite-lattice model will never be able to perfectly simulate the ideal Ising model, which is an infinite lattice.

VII. COMPUTATIONAL TECHNIQUES, REPRODUCIBILITY, AND CODE AVAILABILITY

Environment and Repository. All simulations were run on macOS using Python 3.9.6 with `numpy`, `matplotlib`, and `scipy`. Full setup instructions, simulation parameters, and a figure reproduction guide are documented in the repository README:

<https://github.com/edenmhuang/Physic-77-88-Simulating-2D-Ising-Model>

The two Jupyter notebooks run end-to-end on UC Berkeley Datahub. `ising_data.npz` and `ising_delta_data(2).npz` are included in the repository and can also be re-generated from scratch via `01_simulate_ising.ipynb`.

VIII. CONCLUSIONS AND FUTURE WORK

A simulation of the 2D Ising model can recover universal critical phenomena. One of the limitations with a simulation is that the lattice must be finite. A next step would be to re-run the simulations detailed here on a larger lattice and compare the resulting critical expo-

nents. Another next step would be to re-run the simulations detailed here without biasing the initial conditions and with a longer burn-in phase.

AUTHOR CONTRIBUTIONS

We both contributed to coding the simulation, making visualizations, and writing the final report. Eden ran the simulations for burn-in time and β and set up the GitHub repository. Alex ran the simulation for δ and did the statistical analysis needed to calculate our estimated values of β and δ .

USE OF AI TOOLS (DISCLOSURE)

We used Claude and Gemini for debugging.

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